

Week 2. Security Labs OPenSSL.

Understanding basic security requirement.

Able to configure various security requirement.

Download the OPenSSL for windows from

<http://slproweb.com/products/Win32OpenSSL.html>

You need to install two files into your computer.

[Win32 OpenSSL v1.0.1h Light](#)

And [Visual C++ 2008 Redistributables](#)

1. Symmetric encryption. Google to find the answer

Encryption using symmetric key

2. Decryption using symmetric key. Google to find the answer

Note: you must use the same encryption algorithm to decrypt. There are few type of encryption algorithm such as des, des3, idea, aes etc. and each with various options. In total there are about 58 combinations.

3. Hashing. Google to find the answer

4. Asymmetric encryption RSA.

1st you need to generate the key first. First you need to generate the key pair for each person involved.

First generate the private key

```
openssl genrsa -out testkey.private 2048
```

then from the private key, generate the public key

```
openssl rsa -in testkey.private -pubout -out testkey.public
```

You can use 1024, 2048 or 4096 key length. Longer key will take an exponential time to produce.

You can check the file by

```
C:\OPENSS~1\bin>type testkey.private
```

```
-----BEGIN RSA PRIVATE KEY-----
```

```
MIIEpgIBAAKCAQEAPeFU1aM+jGRGqPqIrpD/fHu3AloBpQ7z2mRptVMJH21JQVG  
z3+0SC3fXzORuAV5XMnx/53Axs/EcT08ab1FRBtJEPxaSBubxTncZ37lgjRR7JA9
```

```
..../* output too long omitted */
```

```
eN6yC2LZx01BvOVc9rDNiWyjIa9MoY5VNxglwsFcubZljq41hQwak4zb
```

```
-----END RSA PRIVATE KEY-----
```

```
C:\OPENSS~1\bin>type testkey.public
```

```
-----BEGIN PUBLIC KEY-----
```

```
MIIBIjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEAPeFU1aM+jGRGqPqIrpD  
/fHu3AloBpQ7z2mRptVMJH21JQVGz3+0SC3fXzORuAV5XMnx/53Axs/EcT08ab1F  
RBtJEPxaSBubxTncZ37lgjRR7JA9cv/hmiwohGSYoUxy/cnb16Jk38ehXNrmtDVa
```

```
..../* output too long omitted */
```

```
OE5A76da8m9IgMRN+lSqyv5qfBOGZzlGgM1B+EeP4Q4k0gPw2DlDyKKEaCu1sePi  
TwIDAQAB
```

```
-----END PUBLIC KEY-----
```

Let said for Bob and Alice and the keys are Bob.priv, Bob.pub, Alice.priv and Alice.pub

```
openssl genrsa -out Bob.priv 2048
openssl rsa -in Bob.priv -pubout -out Bob.pub
openssl genrsa -out Alice.priv 2048
openssl rsa -in Alice.priv -pubout -out Alice.pub
```

Special Note:

openssl rsautl is used to do RSA encryption and decryption
when a public key is used as input –pubin option must be included.

Scenario 1.

Alice want to send an encrypted message to Bob. Only bob can read the message.
The filename is file1.txt

The content of file1.txt is “ For the first time, MMU student boleh encrypt/decrypt”.

C:\OPENSS~1\bin>type file1.txt
For the first time, MMU student boleh encrypt/decrypt

Encryption command. Google to find the answer

Decryption command. Google to find the answer

Scenario 2.

Alice wants to make a public announcement. Bob need to verify that the message is coming from Alice and non-body else. The file is file1.txt

Encryption . Google to find the answer

Decryption Google to find the answer

Quizes: Many different scenario will be given such as how to use OpenSSL to do a digital certificate. DO some googling before quiz ok.

E.g. Scenario 3.

Alice wants to send an encrypted message (file1.txt) which only Bob can decrypt it. At the same time, Bob needs to verify that the message is sent by Alice and not anyone else. Write down the encryption steps and show a working solution to the lecturer using openssl command as shown in Lab 10 Security Lab.

Let said it for Bob and Alice and the keys are Bob.priv, Bob.pub, Alice.priv and Alice.pub. List down the security (encrypt, decrypt, sign) steps done by Alice and Bob.

Possible LAB TEST.

Alice wants to send a message to Bob with the following requirement. There are 16 scenarios with different requirement. Some requirement may not make any senses at all. Please list down all the possible encryption, decryption, hashing steps and command for Bob and Alice.

The initial RSA keys used are Alice.public, Alice.private, Bob.public and Bob.private.

Nevertheless, you can generate other keys such as session key, symmetry keys from the random number generator and the RSA keys.

No	Secrecy	Message Integrity	Non-repudiation	Replay Protection	List down the possible steps for Alice and Bob
1	No	No	No	No	
2	No	No	No	Yes	
3	No	No	Yes	No	
4	No	No	Yes	Yes	
5	No	Yes	No	No	
6	No	Yes	No	Yes	
7	No	Yes	Yes	No	
8	No	Yes	Yes	Yes	
9	Yes	No	No	No	
10	Yes	No	No	Yes	
11	Yes	No	Yes	No	
12	Yes	No	Yes	Yes	
13	Yes	Yes	No	No	
14	Yes	Yes	No	Yes	
15	Yes	Yes	Yes	No	
16	Yes	Yes	Yes	Yes	